

II. Digestive System

1. Which of the following are not the tastes of the tongue?

- | | |
|------------|-----------|
| 1. Sweet | 2. Bitter |
| 3. Salty | 4. Spicy |
| 5. Umami | 6. Sour |
| 7. Pungent | |

Select the correct answer using the codes given below.

- | | |
|----------------|----------------|
| (a) 2, 5 and 7 | (b) 1, 3 and 4 |
| (c) 4 and 7 | (d) 3 and 6 |

69th B.P.S.C. (Pre) 2023

Ans. (c)

Sweet, Bitter, Salty, Umami and Sour are the tastes of the tongue but Tongue can not detect the Spicy and Pungent taste.

2. The sensitive area of the human tongue to bitterness is:

- | | |
|--|-----------------|
| (a) Tip | (b) Middle part |
| (c) Posterior part | (d) Edge |
| (e) None of the above / More than one of the above | |

66th B.P.S.C. (Pre) 2020

Ans. (c)

Sweet, sour, salty, bitter and savory (umami) tastes can actually be sensed by all parts of the human tongue. Only the sides of the tongue are more sensitive than the middle overall. The back of our tongue (i.e. posterior part) is very sensitive to bitter tastes. This is apparently to protect us so that we can spit out spoiled or poisonous foods or substances before they enter the throat and are swallowed.

3. Most of the enzymes are –

- | | |
|--|-------------|
| (a) proteins | (b) lipids |
| (c) acids | (d) alkalis |
| (e) None of the above/More than one of the above | |

67th B.P.S.C. (Re-Exam) (Pre) 2021

Ans. (a)

Enzymes are basically proteins, which are usually formed by the synthesis of nitrogenous substances or amino acids. In biological activity a particular type of enzyme performs a particular function. They work in the same way as a lock has a special key to open it.

4. The enzyme which is mixed with food in our mouth is

- | | |
|-----------------------|--------------------------------|
| (a) pepsin | (b) peroxidase |
| (c) amylase | (d) More than one of the above |
| (e) None of the above | |

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Ans. (c)

The enzyme mixed with food in our mouth is called "Salivary amylase" or "ptyalin". It helps break down starches into smaller sugar molecules like maltose and dextrin during the process of digestion.

5. Substances which are responsible for catalyzing chemical reactions in living systems are -

- | | |
|--------------|--------------|
| (a) Bacteria | (b) DNA |
| (c) Enzymes | (d) Proteins |

44th B.P.S.C. (Pre) 2000

Ans. (c)

Enzymes are biological molecules (proteins) that act as bio-catalysts and helps complex reactions occur in the body. These are basically proteins which accelerate chemical reactions in living systems. Almost all metabolic processes in the cell need enzymes catalysis in order to occur at rates fast enough to sustain life.

6. The source of the enzyme 'lipase' is :

- | | |
|--|--------------|
| (a) Kidney | (b) Pancreas |
| (c) Liver | (d) Heart |
| (e) None of the above/More than one of the above | |

66th B.P.S.C. (Pre) (Re. Exam) 2020

Ans. (b)

The enzyme 'lipase' is produced in the pancreas, mouth and stomach. It is a fat hydrolysing enzyme. Its function is to digest fats and lipids, helping to maintain correct gallbladder function.

7. Energy is stored in liver and muscles in the form of :

- | | |
|------------------|--------------|
| (a) Carbohydrate | (b) Fat |
| (c) Protein | (d) Glycogen |

56th to 59th B.P.S.C. (Pre) 2015

Ans. (d)

Glycogen is the principal storage form of glucose or energy. In humans, it is stored in liver and muscle cells and can be converted to glucose.

8. What does the cell usually use to produce energy?

- | | |
|-----------------------|--------------------------------|
| (a) Amino acid | (b) Glucose |
| (c) Fatty acid | (d) More than one of the above |
| (e) None of the above | |

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Ans. (b)

The food that is usually utilized by the cells is in the form of glucose. The glucose is oxidized to produce carbon dioxide and water with the release of energy and this process is called cellular respiration. Here option (b) is the correct answer.